DS605: Fundamentals of Machine Learning

Lab Assignment - 2

Vectorized Programming with NumPy and Data Wrangling with Pandas

Dataset (Part B): Kaggle Titanic dataset (train.csv)
Submission: Public GitHub repository link only.

Objective

Practice vectorized NumPy operations and basic data wrangling with Pandas using the Titanic dataset.

Part A - Vectorized Programming with NumPy

Task 1 - Arrays, Statistics, and Indexing

- Generate Array A with 100 random integers. Set a random seed for reproducibility.
- Use NumPy functions to find min, max, median, mean, and standard deviation of Array A.
- Generate Array B with exactly 100 values using np.arange().
- Implement np.zeros() and np.ones(). Display shape and data type.
- Implement np.linspace(). Briefly state how it differs from np.arange().
- Create one 2D and one 3D array. Show shape, ndim, indexing, rows, columns, and slicing.
- Use reshape() to create a matrix, then flatten it back to 1D.

Task 2 - Vectorized Arithmetic and Linear Algebra

- Create two matrices. Perform addition, element-wise multiplication, and matrix multiplication using @ or np.matmul().
- For a square matrix, find its transpose and determinant.
- If invertible, find the inverse and verify A @ A^-1 using np.allclose().
- Use vectorized NumPy operations; avoid explicit Python loops.

Task 3 - Normal Distribution and Histogram

- Generate at least 1,000 values from a normal distribution. State the chosen mean and standard deviation.
- Find the sample mean and sample standard deviation. Compare them with the chosen values.
- Plot a histogram with it.

Part B - Data Wrangling with Pandas: Titanic Dataset

Task 4 - Load and Inspect Data

- Load train.csv. Implement head(), tail(), shape, columns, info(), and describe().
- Use loc and iloc to select rows/columns. Give one example of each and state the difference.

Task 5 - Filtering and Querying

Use Boolean indexing and/or query() to answer:

- Male passengers older than 50: how many?
- Female first-class passengers: how many, and what percentage survived?
- Age 20-40, Fare above overall median, and survived: how many?
- Travelling alone (SibSp=0, Parch=0), age < 30, and did not survive: how many?
- Embarked='S', Pclass 2 or 3, and Fare above Southampton median: how many?

Task 6 - groupby() and Aggregation

- Survival rate by Sex.
- Survival rate by Pclass.
- Average Age and Fare by Pclass.
- Passenger count and survival rate by Sex-Pclass.
- Passenger count, average Fare, and survival rate by Embarked.

Task 7 - Missing Values and Fare Outliers

- Find the missing-value count and percentage for every column. Plot missing counts as a bar chart.
- Fill missing Age values with the mean Age. Show missing counts before and after.
- Try to fill NaN values by imputing using: mean, median, mode, random value
- For Fare, calculate Q1, Q3, IQR, 1.5xIQR bounds, and count the outliers.

Task 8 - Features and Pivot Table

- Create FamilySize = SibSp + Parch + 1 and IsAlone = 1 when FamilySize = 1.
- Create a pivot table: rows=Sex, columns=Pclass, values=mean Survived. Find the highest and lowest groups.

Task 9 - Visualizations and Observations

- Plot a correlation heatmap for relevant numerical columns. Note the strongest positive and negative relationships.
- Plot the survival rate by Sex and report the male/female survival rates.
- Plot Age vs Fare and distinguish Survived=0 and Survived=1.
- Write 5-7 short observations supported by your numerical results or plots.

Deliverables (GitHub Only)

- README with assignment title, name, ID, dataset, project details, and key observations.
- Complete runnable NumPy/Pandas code (Jupyter Notebook recommended).
- Cleaned Titanic dataset, generated figures, and final observations. Submit only the public GitHub link.